Quantization of Constrained Systems

The Abelian Higgs Model as a Case Study

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Abstract

[Abstract, to be written last. Planned content: the companion note *Toric Code Demystified* builds its lattice model on temporal gauge, Gauss's law imposed on states, and compact link variables. This note examines these choices in the continuum abelian Higgs model: whether fixing one gauge condition or two matters classically (the drift of the Coulomb condition, the extended Hamiltonian, the identity $\partial_t(\nabla\cdot A) - \nabla^2 A_0^{\text{eff}} = \rho$) and quantum mechanically (both give the same theory), why the lattice nevertheless fixes only one (locality), what unitary gauge does and does not say about the Higgs phase, and how compactness leads to charge quantization, holonomies and the toric code.]

Companion note: *Toric Code Demystified* (lattice gauge theory, $\mathbb{Z}_2$ reduction, and topological order).

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1 Introduction

1.1 What the companion note takes for granted

To write (one page). The lattice construction of *Toric Code Demystified* rests on three choices that are made there without discussion: (i) temporal gauge, i.e. fixing only one of the two first-class constraints; (ii) Gauss's law imposed on states rather than solved; (iii) compact link variables, so that electric flux and charge are integers. This note examines each choice in the continuum abelian Higgs model and asks what would change if it were made differently. The guiding question: fixing one gauge condition or two — does it matter, and if not, why does the lattice insist on one?

1.2 Plan and main results

To write: a short table, one line per section. §2: constraints and temporal gauge (classical); §3: one condition or two — the drift, the extended Hamiltonian, the gauge-invariant identity $\partial_t(\nabla\cdot A)-\nabla^2 A_0^{\text{eff}} = \rho$, and the quantum statement that both give the same theory; §4: why the lattice nevertheless fixes only one (locality of the reduced algebra); §5: unitary gauge and the Higgs phase (what is gauge invariant); §6: compactness, charge quantization and holonomies, leading to the toric code; Appendix A: constraint chains and depth.

Conventions. State once: $\vec{E} = \dot{\vec{A}} - \nabla A_0$, $H_c \ni -\int A_0\Phi_2$, $\{A_i(x), E_j(y)\} = \delta_{ij}\delta^3(x-y)$, $\Phi_1 = \pi^0$, $\Phi_2 = \nabla\cdot E - \rho$; units $\hbar = c = 1$.

2 Constrained dynamics of the abelian Higgs model

2.1 Lagrangian and primary constraint

The abelian Higgs model — equivalently scalar electrodynamics, or relativistic Ginzburg–Landau — is

$$\mathcal{L} = -\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} + \big(D_\mu\phi\big)^*\big(D^\mu\phi\big) - V\big(|\phi|^2\big), \qquad D_\mu = \partial_\mu - ieA_\mu. \tag{1}$$

In three-vector notation, with $\vec{E} = \dot{\vec{A}} - \nabla A_0$ and $\vec{B} = \nabla\times\vec{A}$,

$$\mathcal{L} = \tfrac{1}{2}\big(E^2 - B^2\big) + \big|D_0\phi\big|^2 - \big|\vec{D}\phi\big|^2 - V. \tag{2}$$

The covariant derivative acts in the time direction as well, $D_0\phi = \dot{\phi} - ieA_0\phi$, and it is worth following where that A_0 ends up: not in the energy, but as the coefficient of a constraint.

2.1.1 Momenta

$$\pi_i = \frac{\partial \mathcal{L}}{\partial \dot{A}_i} = E_i, \qquad \pi_\phi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \big(D_0\phi\big)^*, \qquad \boxed{\pi^0 = \frac{\partial \mathcal{L}}{\partial \dot{A}_0} = 0} \tag{3}$$

The last is not an equation to be solved but a *primary constraint*,

$$\Phi_1 := \pi^0 \approx 0, \tag{4}$$

and it arises because $F_{\mu\nu}$ is antisymmetric, so $\dot{A}_0$ never appears in (1).

2.1.2 The Legendre transform of a singular Lagrangian

Remark 2.1 (why a multiplier appears)**.** The Legendre transform is performed as usual,

$$H = \pi^0 \dot{A}_0 + \pi^i \dot{A}_i + \pi_\phi \dot{\phi} + \text{c.c.} - \mathcal{L}, \tag{5}$$

but the map $\dot{q} \mapsto p$ is not invertible: the relations for π^i, π_ϕ can be solved for $\dot{A}_i, \dot{\phi}$, whereas $\pi^0 = 0$ carries no information about $\dot{A}_0$. Since $\mathcal{L}$ contains no $\dot{A}_0$, all remaining terms are independent of it, so

$$H = \underbrace{\big[\pi^i \dot{A}_i + \pi_\phi \dot{\phi} + \text{c.c.} - \mathcal{L}\big]}_{H_c(q,p)} + \dot{A}_0\, \pi^0, \tag{6}$$

and $\dot{A}_0$ is simply not determined by the phase-space data. Renaming it $\lambda_1 := \dot{A}_0$ gives the total Hamiltonian. The term is not inserted by hand; it is the $\pi^0 \dot{A}_0$ term of the Legendre transform, with the undetermined velocity kept as an unknown. Consistency is immediate: $\dot{A}_0 = \{A_0, H_T\} = \lambda_1$.

$$H_c = \int \mathrm{d}^3x \, \Big[\underbrace{|\pi_\phi|^2 + \big|\vec{D}\phi\big|^2 + V}_{\text{matter}} + \underbrace{\tfrac{1}{2}\big(E^2 + B^2\big)}_{\text{gauge field}} \Big] \; - \; \int \mathrm{d}^3x \; A_0 \, \big(\nabla\cdot E - \rho\big), \tag{7}$$

$$H_T = H_c + \int \mathrm{d}^3x \; \lambda_1 \, \pi^0, \qquad \rho = ie\big(\phi\, \pi_\phi - \phi^* \pi_\phi^*\big). \tag{8}$$

2.2 The second constraint, and why both are first class

Proposition 2.2. *Consistency of Φ_1 produces the Gauss constraint*

$$\Phi_2 := \nabla\cdot E - \rho \approx 0, \tag{9}$$

and the procedure terminates there.

Proof. $\dot{\Phi}_1 = \{\pi^0, H_T\} = -\,\partial H_T / \partial A_0 = \nabla\cdot E - \rho$. That $\dot{\Phi}_2 = 0$ identically is Proposition 2.3 below. □

Proposition 2.3. *Φ_1 and Φ_2 are first class, and $\{\Phi_2, H_T\} = 0$ identically.*

Proof. Smear the Gauss constraint, $G[\alpha] = \int \alpha\, \Phi_2$. Reading off its Poisson brackets (integrating $\int \alpha\, \partial_j E_j$ by parts),

$$\{A_i, G\} = -\partial_i \alpha, \quad \{E_i, G\} = 0, \quad \{\phi, G\} = -ie\,\alpha\phi, \quad \{\pi_\phi, G\} = +ie\,\alpha\pi_\phi, \tag{10}$$

so $G[\alpha]$ generates precisely a $U(1)$ gauge transformation. Under it

$$\delta\big(D_i\phi\big) = \partial_i(\delta\phi) - ie(\delta A_i)\phi - ieA_i\,\delta\phi = -ie\,\alpha\, D_i\phi, \tag{11}$$

which is covariance, so $|D_i\phi|^2$ is invariant; $|\pi_\phi|^2$ is invariant because ϕ and π_ϕ carry opposite phases; $V(|\phi|^2)$ because $|\phi|^2$ is inert; E because $\delta E = 0$; and B because $\delta B = \nabla \times (-\nabla\alpha) = 0$. Hence the first integral of (7) has vanishing bracket with G. For the second, A_0 commutes with everything in G, and $\{\Phi_2(x), \Phi_2(y)\} = 0$ since the only potentially nonzero piece, $\{\rho(x), \rho(y)\}$, is proportional to $-\phi(x)\pi_\phi(y)\delta + \phi(y)\pi_\phi(x)\delta$, which vanishes at $x = y$. Finally $\{\Phi_1, \Phi_2\} = 0$ because Φ_2 contains no A_0. □

Remark 2.4. Two first-class constraints remove $2 \times 2 = 4$ from the eight phase-space dimensions of (A_μ, π^μ), leaving four: two polarizations. Each first-class constraint removes two dimensions, one for the constraint and one for the gauge orbit it generates.

A two-variable model with the same constraint structure — one gauge field, one Gauss law, one multiplier — is set up in §3.1, where it is used as a sign-check device.

2.3 Fixing one constraint: temporal gauge

Impose the single gauge condition

$$\chi_1 := A_0 \approx 0. \tag{12}$$

Since $\{\Phi_1, \chi_1\} = -1$, the pair (Φ_1, χ_1) is second class and (π^0, A_0) may be eliminated.

Proposition 2.5. *Eliminating (π^0, A_0) changes no bracket among the remaining variables.*

Proof. For F, G built from $(A_i, E_i, \phi, \pi_\phi)$ one has $\{F, \pi^0\} = \{F, A_0\} = 0$, so the Dirac correction $\{F, \Xi_a\}C^{ab}\{\Xi_b, G\}$ vanishes term by term. □

What vanishes is the correction term, not the difference between the two brackets: $\{A_0, \pi^0\}_D = 1-1 = 0$, the eliminated pair becoming Casimir. The precise statement is that $C^\infty(A_i, E_i, \phi, \pi_\phi)$ is the Poisson commutant of $\{\Phi_1, \chi_1\}$, which by Proposition 4.3 is what allows the ambient bracket to compute the reduced one — and, by Interpretation 4.4, is the property that makes this elimination behave better than the alternative.

Setting $A_0 = 0$ in (7) deletes the last integral, leaving the familiar

$$\boxed{H = \int \mathrm{d}^3x \left[|\pi_\phi|^2 + \left|\vec{D}\phi\right|^2 + V + \tfrac{1}{2}\left(E^2 + B^2\right) \right]} \tag{13}$$

with Φ_2 still first class and therefore imposed on states,

$$\left(\nabla\cdot\hat{E} - \hat{\rho}\right)|\psi\rangle = 0. \tag{14}$$

The residual gauge freedom is a time-independent $\alpha(\vec{x})$, generated by Φ_2.

Remark 2.6. The phase-space variable A_0 has no tendency to move: $\dot{A}_0 = \lambda_1$, and $\lambda_1 = 0$ keeps $A_0 = 0$ for all time. Nothing in the dynamics pushes it. In particular Φ_2 cannot move it, since $\{A_0, \Phi_2\} = 0$.

This is the form one discretizes, because it is local and because (14) becomes a condition at each lattice site.

2.4 A question, and what is deferred

Everything the lattice construction of the companion note *Toric Code Demystified* needs is now in place: (13) with (14) imposed on states, and Proposition 2.5 guaranteeing that the canonical bracket survives the elimination of (π^0, A_0) intact.

One question should be raised before leaving, because it is easy to get wrong and because the way it is usually asked is not well posed. Temporal gauge sets $A_0 = 0$; Coulomb gauge

sets $\nabla\cdot A = 0$. Each is standard, each is available, and it is tempting to use both at once — the combination is even given a name, radiation gauge. Can they hold together?

The answer is developed in §3. Two of its results are worth recording here. *First*, the field playing the role of the Lagrangian A_0 in the equations of motion is not the phase-space coordinate but

$$A_0^{\text{eff}} := A_0 - \lambda_2, \tag{15}$$

λ_2 being the multiplier that the extended Hamiltonian attaches to Φ_2; under H_T, where no such multiplier exists, $A_0^{\text{eff}} = A_0$ (§3.3.1). *Second*, in terms of it the entire content of the question is one identity (§3.4):

$$\partial_t\big(\nabla\cdot A\big) - \nabla^2 A_0^{\text{eff}} = \rho. \tag{16}$$

Setting either term to zero determines the other, so the two conditions are compatible only when $\rho = 0$. The temporal form used below is the case $A_0^{\text{eff}} = 0$, with Gauss's law surviving as (14) rather than being solved. Nothing in the lattice construction depends on any of this.

Remark 2.7 (what to discretize)**.** The temporal form (13) is the one to put on a lattice: it is local, and the surviving first-class constraint becomes a condition at each site. Writing $\phi_v = \mathrm{e}^{i\theta_v}$ with conjugate integer charge n_v, and A_ℓ on links with conjugate E_ℓ, Gauss's law becomes the discrete divergence

$$\sum_{\ell\ni v} E_\ell = n_v, \tag{17}$$

and for gauge group $\mathbb{Z}_2$, exponentiating by $\mathrm{e}^{i\pi(\cdot)}$ turns this into $\prod_{\ell\ni v}\sigma_\ell^x = \tau_v^x$, the familiar vertex condition of lattice $\mathbb{Z}_2$ gauge theory with matter. The Coulomb form (29), derived in §3, is unsuitable for that purpose, being nonlocal.

3 One gauge condition or two

Section 2 stops where the lattice construction of the companion note stops needing it, leaving the question of §2.4 unanswered: can $A_0 = 0$ and $\nabla\cdot A = 0$ hold at once? This section answers it, and then follows the two questions the answer raises.

§3.1–§3.4 settle the original question. A two-variable model fixes every sign; the Coulomb condition is imposed and develops a drift; the extended Hamiltonian is what imposing both conditions literally requires; and the invariant content turns out to be the single identity of Theorem 3.22, written in terms of $A_0^{\text{eff}} = A_0 - \lambda_2$ rather than A_0.

That answer rests on a count — one arbitrary function in H_T against two gauge conditions — which looks like a shortage of freedom. §A.1 shows it is not one, for any number of constraints, and in doing so produces the notion that does distinguish the two gauge conditions: their depth in the constraint chain. But depth is measured against a chain anchored at the primary constraints, so it is only as invariant as that anchoring is. §4.1 asks whether anything about the difference between Φ_1 and Φ_2 survives that objection, and finds that what does is neither of the two classifications one is usually offered.

[Revise this introduction for the new layout. This section now carries the classical question (§3.1–§3.4), a summary of why one arbitrary function suffices (§3.5; proofs in Appendix A), and the quantum comparison (§3.6): fixing one condition and imposing Gauss's law on states (Q1), or fixing two and quantizing the Dirac bracket (Q2), give the same quantum theory. The asymmetry between Φ_1 and Φ_2 is now §4.]

3.1 A model with the same structure

Everything below can be followed in a system with two degrees of freedom. Let

$$L = \tfrac{1}{2}\big(\dot{q} - a\big)^2 + a\,\rho, \qquad \rho = \text{const}, \tag{18}$$

with the dictionary $q \leftrightarrow A_i$, $a \leftrightarrow -A_0$, $p_q \leftrightarrow E$, $\rho \leftrightarrow$ charge density — the sign is not optional, see Remark 3.1. Then

$$p_q = \dot{q} - a, \qquad p_a = 0 =: \Phi_1, \qquad H_c = \tfrac{1}{2}p_q^2 + a\,p_q - a\rho, \tag{19}$$

and $\dot{\Phi}_1 = -\partial H_c/\partial a = \rho - p_q$ gives

$$\Phi_2 := p_q - \rho \approx 0, \tag{20}$$

the Gauss constraint. The gauge transformation is $\delta q = \epsilon$, $\delta a = \dot{\epsilon}$, under which (18) changes by a total derivative. The equation of motion is

$$\dot{q} = \frac{\partial H_T}{\partial p_q} = p_q + a. \tag{21}$$

Remark 3.1 (the dictionary, with signs; the model as a sign-check device)**.** The model has no spatial structure, so ∇ and $\nabla\cdot$ both collapse to numbers — but not to $+1$ for both, since adjointness $\nabla\cdot\ = -\nabla^{\mathsf{T}}$ has to survive. Comparing one bracket fixes the assignment uniquely:

$$\{\chi_2, \Phi_2\}\Big|_{\text{toy}} = \{q, p_q\} = +1 \qquad \text{against} \qquad \{\chi_2, \Phi_2\}\Big|_{\text{field}} = -\nabla^2 \quad\Longrightarrow\quad \nabla^2 \longmapsto -1, \tag{22}$$

whence

$$\nabla\cdot \;\longmapsto\; +1, \qquad \nabla \;\longmapsto\; -1, \qquad \nabla^2 \;\longmapsto\; -1, \qquad a \;\longleftrightarrow\; \nabla A_0 = -A_0. \tag{23}$$

The last identification is forced independently by the Hamiltonians: here $H_c = \frac{1}{2}p_q^2 + a\,\Phi_2$, against $H_c \ni -\int A_0\,\Phi_2$ in (7). With (23) in hand the model is a reliable check on every sign in §3; the value $\lambda_2 = -\rho$ obtained in §3.3 and Example 3.16 is one instance.

3.2 Fixing both: the Coulomb condition and where it drifts

Now add

$$\chi_2 := \nabla\cdot A \approx 0, \qquad \{\chi_2, \Phi_2\} = -\nabla^2, \tag{24}$$

which is invertible away from the zero mode. The bracket matrix of $(\Phi_1, \Phi_2, \chi_1, \chi_2)$ is block diagonal and of full rank apart from that single mode (the constant α, i.e. the total charge), so the four are second class and Dirac brackets exist.

3.2.1 The drift, and its cause

Proposition 3.2. *With the Hamiltonian* (8)*,*

$$\{\chi_2, H_T\} = \nabla\cdot E \;+\; \nabla^2 A_0. \tag{25}$$

Proof. $\{A_i, \frac{1}{2}\int E^2\} = E_i$ and $\{A_i, -\int A_0\,\nabla\cdot E\} = \partial_i A_0$ after an integration by parts, so $\{A_i, H_T\} = E_i + \partial_i A_0$ with *both* signs positive — as they must be, that relation read backwards being $\vec{E} = \dot{\vec{A}} - \nabla A_0$. Take the divergence. (Verified numerically on a discrete lattice.) □

Equation (25) contains the whole story.

- Demanding $\dot{\chi}_2 = 0$ *determines* A_0: $\nabla^2 A_0 = -\nabla\cdot E = -\rho$, i.e. $A_0 = \frac{1}{4\pi}\int \rho(y)/|x-y|\,\mathrm{d}^3y$. This is the Coulomb potential — with the sign that makes it *the* Coulomb potential, and it is the second, usually unwritten, condition of the Coulomb gauge. No modification of H_T is involved: the Hamiltonian is untouched and a phase-space variable is fixed (Proposition A.8).
- If instead $A_0 = 0$ has already been imposed, then $\dot{\chi}_2 = \nabla\cdot E = \rho \neq 0$ and the Coulomb condition is not preserved.

So both gauge conditions are, in the end, conditions on A_0, and they ask for different values.

Remark 3.3 (the two named gauges)**.** *Temporal gauge* fixes χ_1 only; it is a deliberately incomplete gauge fixing, and Φ_2 survives as (14). *Coulomb gauge* fixes both, with $\chi_1 : A_0 + \nabla^{-2}\rho$ and $\chi_2 : \nabla\cdot A$; it is complete, and — a point developed in §A.1, Proposition A.8 — needs no extension of H_T whatsoever. The combination $A_0 = 0$ together with $\nabla\cdot A = 0$ is *radiation gauge*, and it exists only when $\rho = 0$.

Remark 3.4 (the drift as a curvature)**.** To write (half a page). Gauss's law is the statement that the motion is orthogonal to the gauge orbits in the kinetic metric, i.e. horizontal for the mechanical connection $\mathcal{A} = I^{-1}\langle\xi,\cdot\rangle$ with locked inertia $I = -\nabla^2 + e^2|\phi|^2$. With charged matter this connection is curved, so no gauge slice can be horizontal (Frobenius), which is why a Coulomb condition imposed in temporal gauge drifts. Zero-dimensional model: $Q = \mathbb{R}_a \times \mathbb{C}$, $J = \dot{a} + er^2\dot{\theta}$, $I = 1 + e^2r^2$; on $J = 0$, $da = -\frac{er^2}{1+e^2r^2}d\chi$ with $\chi = \theta - ea$, curvature $-\frac{2er}{(1+e^2r^2)^2}\,dr\wedge d\chi$, vanishing for $e = 0$. [Check C8.]

3.2.2 The Dirac bracket removes the drift

Proposition 3.5. *Once* (Φ_2, χ_2) *is also taken into account, the drift disappears:* $\{\chi_2, H\}_D = 0$*, and the flow becomes transverse,*

$$\{A_i, H\}_D = E_i - \left(\nabla\frac{1}{\nabla^2}\nabla\cdot\right)_{ij} E_j = E_i^T. \tag{26}$$

Proof. For any second-class Ξ_c and any G, $\{\Xi_c, G\}_D = \{\Xi_c, G\} - \{\Xi_c, \Xi_a\}C^{ab}\{\Xi_b, G\} = 0$, since $\{\Xi_c, \Xi_a\}C^{ab} = \delta_c^b$. Applying the same formula to A_i produces (26); the subtracted piece is the longitudinal projector. (Both verified numerically.) □

Remark 3.6 (geometry)**.** The constraint surface $C = \{\Phi_2 = 0\}$ is coisotropic: $\omega|_C$ is degenerate, its kernel being the gauge orbits. A gauge condition selects a section $\Sigma = C \cap \{\chi_2 = 0\}$ meeting each orbit once, and $\omega|_\Sigma$ is nondegenerate — that is the Dirac bracket. The Hamiltonian vector field X_H is tangent to C but *not* to Σ, so the induced flow on Σ is X_H projected along the orbits,

$$X_H^\Sigma = X_H + X_{\text{gauge}}(\alpha), \qquad \nabla^2\alpha = \rho. \tag{27}$$

The compensating parameter α is exactly minus the Coulomb potential, i.e. the multiplier λ_2 of Proposition 3.10. Nothing is inconsistent here; a section simply has to be dragged along to stay a section.

[Expand Remark 3.6 into a proposition with proof, since it is the one general fact the quantum comparison of §3.6 rests on: for second-class Ξ_a, $\{f,g\}_D|_\Sigma$ is the Poisson bracket of $\omega|_\Sigma$. Proof via $T\Sigma \oplus T\Sigma^\omega$ and $PX_f = X_f - c^a X_{\Xi_a}$. One finite-dimensional illustration (particle on S^2, checked) as an example environment below.]

Example 3.7 (the Dirac bracket of a particle on S^2)**.** $\Xi = (x^2 - 1, x\cdot p)$, $C_{12} = 2x^2$; $\{x_i, p_j\}_D = \delta_{ij} - x_ix_j/x^2$, $\{p_i, p_j\}_D = (x_jp_i - x_ip_j)/x^2$, which on Σ equal the canonical brackets of T^*S^2. [Check C1.]

3.2.3 Eliminating the longitudinal field

Since Φ_2 is now second class, E_L is not independent:

$$\nabla\cdot E_L = \rho \quad\Longrightarrow\quad \tfrac{1}{2}\int E_L^2 = \frac{1}{2}\int \rho\,\frac{1}{-\nabla^2}\,\rho = \frac{1}{8\pi}\iint \frac{\rho(x)\rho(y)}{|x-y|}, \tag{28}$$

verified numerically. The reduced Hamiltonian is

$$H = \int \Big[|\pi_\phi|^2 + \big|\vec{D}\phi\big|^2 + V + \tfrac{1}{2}\big(E_T^2 + B^2\big)\Big] + \frac{1}{8\pi}\iint \frac{\rho\,\rho}{|x-y|}. \tag{29}$$

The longitudinal electric field has not disappeared; it has turned into the instantaneous Coulomb interaction.

Remark 3.8 (why no retarded ambiguity)**.** Gauss's law is an *elliptic* equation on a time slice, not a hyperbolic equation on spacetime, so it has a unique solution given decay at spatial infinity. Retarded and advanced Green functions are alternatives for $\Box A = j$, which governs the transverse sector; the longitudinal sector carries no propagating degree of freedom and hence no choice. The instantaneous piece is gauge dependent and cancels against transverse exchange in observables.

3.3 Imposing both conditions literally: the extended Hamiltonian

Section 2.3 showed that A_0 can be held at zero forever, and the previous section showed that $\nabla\cdot A$ can be held at zero forever. Can both be done at once? The answer requires enlarging the Hamiltonian.

Definition 3.9. The *extended Hamiltonian* attaches a multiplier to every first-class constraint, secondary ones included:

$$H_E = H_c + \int \lambda_1\Phi_1 + \int \lambda_2\Phi_2. \tag{30}$$

Proposition 3.10. *With* (30) *the choices* $\lambda_1 = 0$ *and* $\nabla^2\lambda_2 = \rho$ *preserve* both $A_0 = 0$ *and* $\nabla\cdot A = 0$ *for all time.*

Proof. $\dot{A}_0 = \{A_0, H_E\} = \lambda_1 = 0$, since $\{A_0, \Phi_2\} = 0$. And $\dot{A}_i = \{A_i, H_E\} = E_i + \partial_i A_0 - \partial_i\lambda_2$, whose divergence is $\nabla\cdot E + \nabla^2 A_0 - \nabla^2\lambda_2$ by (25), so with $A_0 = 0$

$$\partial_t\big(\nabla\cdot A\big) = \nabla\cdot E - \nabla^2\lambda_2 = \rho - \nabla^2\lambda_2, \tag{31}$$

which vanishes for the stated λ_2. □

Example 3.11 (in the model of §3.1)**.** $H_E = \frac{1}{2}p_q^2 + a p_q - a\rho + \lambda_1 p_a + \lambda_2(p_q - \rho)$ gives $\dot{a} = \lambda_1$ and $\dot{q} = p_q + a + \lambda_2$. On the constraint $p_q = \rho$, choosing $\lambda_1 = 0$ and $\lambda_2 = -\rho$ yields $a = 0$ and $\dot{q} = 0$ simultaneously.

3.3.1 The price

Proposition 3.12. *The total Hamiltonian* H_T *reproduces the Lagrangian relation between velocity and momentum, whereas* H_E *does not:*

$$\{A_i, H_T\} = E_i + \partial_i A_0, \qquad \{A_i, H_E\} = E_i + \partial_i\big(A_0 - \lambda_2\big). \tag{32}$$

Comparing with $\vec{E} = \dot{\vec{A}} - \nabla A_0$, the object playing the role of the Lagrangian A_0 is therefore not the phase-space coordinate but

$$\boxed{A_0^{\text{eff}} := A_0 - \lambda_2} \tag{33}$$

Caution 3.13 (A_0^{eff} is not a second field)**.** There is one field A_0, a coordinate on phase space. What (33) records is which combination appears in the dynamics, and the answer depends on the Hamiltonian: under H_T the coefficient of Φ_2 in H_c is A_0 itself and no multiplier λ_2 exists, so $A_0^{\text{eff}} = A_0$ and the Lagrangian relation $\vec{E} = \dot{\vec{A}} - \nabla A_0$ holds verbatim. Under H_E the coefficient is $A_0 - \lambda_2$, and it is that combination, not A_0, that the equations of motion see. Neither A_0 nor λ_2 is separately meaningful there.

Remark 3.14 (nothing is gained)**.** In the model of §3.1 the solution labelled $a = 0, \lambda_2 = -\rho$ and the solution labelled $a = -\rho, \lambda_2 = 0$ are the same history, $q(t) = \text{const}$; only the bookkeeping differs. Numerically, both give $q = (0, 0, 0, 0)$ on a sample time grid, whereas genuine temporal gauge, $a_{\text{eff}} = 0$, gives $q = (0, 1.7, 3.4, 5.1)$ for $\rho = 1.7$. Dirac's extended Hamiltonian is physically equivalent to H_T; what it buys is a relabelling, at the cost of severing A_0 from the velocity.

3.3.2 Complete gauge fixing, and the counting of multipliers

Both gauge conditions can be imposed at once. When they are, no arbitrary function survives.

Proposition 3.15 (all multipliers are determined)**.** *Impose gauge conditions χ_1, χ_2 with $C_{ab} := \{\chi_a, \Phi_b\}$ invertible. Then consistency of the gauge conditions,*

$$\dot{\chi}_a = \{\chi_a, H_c\} + \sum_b \lambda_b \{\chi_a, \Phi_b\} = 0, \tag{34}$$

has the unique solution

$$\lambda_b = -\sum_a \left(C^{-1}\right)_{ba} \{\chi_a, H_c\}. \tag{35}$$

Proof. The system is linear in the λ_b with coefficient matrix C_{ab}, invertible by hypothesis. □

Example 3.16 (Maxwell)**.** For $\chi_1 = A_0$ and $\chi_2 = \nabla \cdot A$ the matrix is block diagonal, $\{\chi_1, \Phi_1\} = +1$, $\{\chi_2, \Phi_2\} = -\nabla^2$, and (35) gives

$$\lambda_1 = 0, \qquad \nabla^2 \lambda_2 = \rho \quad \Longrightarrow \quad \lambda_2 = -\frac{1}{4\pi} \int \frac{\rho(y)}{|x - y|} \, \mathrm{d}^3 y, \tag{36}$$

which is Proposition 3.10. In the model of §3.1, with $\chi_1 = a$ and $\chi_2 = q - q_0$, the same computation gives $\lambda_1 = 0$ and $\lambda_2 = -\rho$.

Remark 3.17 (Dirac brackets make the multipliers irrelevant)**.** Once all four constraints are second class one may instead pass to Dirac brackets, in which $\{F, \Xi_a\}_D = 0$ for every constraint and hence

$$\{F, H_E\}_D = \{F, H_c\}_D : \tag{37}$$

the multiplier terms drop out entirely. The two procedures agree: solving (35) and substituting, or discarding the multipliers and using $\{\ ,\ \}_D$, produce the same trajectory. This is the reconciliation of §3.3 with §2.3.

How many conditions may be imposed. The count differs between the two Hamiltonians, and this is what underlies everything in §2.3–§3.3. The general form of this count, for N first-class constraints organised into chains, is Theorem A.3; the conclusion there is that the number of resources always equals the number of conditions, so that what follows should be read as a statement about the *kinds* of resource available, not about their number.

Proposition 3.18. *H_T contains exactly one arbitrary function, λ_1; the coefficient of Φ_2 is the phase-space variable A_0, not a multiplier. H_E contains two, λ_1 and λ_2.*

Proof. Immediate from (8) and (30), together with $H_c \ni -\int A_0\Phi_2$ from (7). □

Corollary 3.19. *Under H_T only one gauge condition may be imposed freely. A second condition does not fix a multiplier — there is none left — but determines the* variable A_0*, which by Lemma A.1 is the resource y_1 conjugate to Φ_1; "freely" therefore means "without fixing a phase-space coordinate", not "at all". Concretely, by Proposition 3.2,*

$$\dot{\chi}_2 = \nabla\cdot E \;+\; \nabla^2 A_0 = 0 \qquad \Longrightarrow \qquad \nabla^2 A_0 = -\rho, \tag{38}$$

so χ_1 cannot also be A_0 unless $\rho = 0$.

Interpretation 3.20 (why the counts differ)**.** A Lagrangian gauge transformation is a single arbitrary function, $\delta A_\mu = \partial_\mu \epsilon$, i.e.

$$\delta A_0 = \dot{\epsilon}, \qquad \delta A_i = \partial_i \epsilon. \tag{39}$$

In the Hamiltonian formulation Φ_1 generates the first and Φ_2 the second, but the two are not independent: they are $\dot{\epsilon}$ and ϵ for the *same* ϵ. The total Hamiltonian respects this, carrying one free function; the extended Hamiltonian treats λ_1 and λ_2 as independent and therefore generates a larger set of transformations than the Lagrangian does.

The additional transformations are harmless in the sense that they act trivially on gauge-invariant quantities — if $\{F, \Phi_2\} \approx 0$ then $\{F, H_E\} \approx \{F, H_T\}$, so for instance $\vec{E}$ evolves identically under both — which is the content of Dirac's conjecture. They are not harmless for the bookkeeping: they are exactly the extra freedom that allows $A_0 = 0$ and $\nabla\cdot A = 0$ to be imposed together, and exactly the reason that doing so is a relabelling rather than a new solution.

Remark 3.21 (the whole of §2.3–§3.3 in one line)**.**

$$\underbrace{\text{one gauge function } \epsilon}_{\text{Lagrangian, } H_T} \qquad \text{versus} \qquad \underbrace{\text{two, } \lambda_1 \text{ and } \lambda_2}_{H_E}.$$

Under H_T one condition may be chosen and the other is forced; under H_E both may be chosen, at the cost of severing A_0 from the velocity. Either way the gauge-invariant content is the identity of Theorem 3.22.

3.4 The gauge-invariant statement

We can now say precisely what is and is not a fact about the theory.

Theorem 3.22. *Along any solution, with A_0^{eff} defined by (33),*

$$\boxed{\partial_t\big(\nabla\cdot A\big) - \nabla^2 A_0^{\text{eff}} = \rho} \tag{40}$$

Consequently

(i) $\nabla\cdot A = 0$ for all t forces $\nabla^2 A_0^{\text{eff}} = -\rho$, so $A_0^{\text{eff}} \neq 0$ whenever charges are present;

(ii) $A_0^{\text{eff}} = 0$ forces $\partial_t(\nabla\cdot A) = \rho \neq 0$, so the Coulomb condition cannot hold at all times;

(iii) both simultaneously force $\rho = 0$.

Proof. Equation (40) is Gauss's law $\nabla\cdot E = \rho$ combined with $\vec{E} = \dot{\vec{A}} - \nabla A_0^{\text{eff}}$, which is Proposition 3.12. The three consequences follow by setting the indicated terms to zero. (Verified numerically for a discrete charge distribution.) □

Corollary 3.23. *The question "can $A_0 = 0$ and $\nabla\cdot A = 0$ hold together?" is not gauge invariant as posed. Under H_T the symbol A_0 means A_0^{eff} and the answer is no; under H_E the symbol A_0 is a coordinate decoupled from the dynamics, $A_0^{\text{eff}} = -\lambda_2$, and the answer is yes. The invariant content is entirely in Theorem 3.22.*

3.4.1 Summary

	fixes	A_0^{eff}	Gauss law
temporal	χ_1 only	0	imposed on states, (14)
Coulomb	χ_1, χ_2	$\nabla^{-2}(-\rho)$	solved; Coulomb term in H
radiation	χ_1, χ_2 with $A_0^{\text{eff}} = 0$	0	only if $\rho = 0$

object	status	value
A_0 (phase-space coordinate)	label; drops out after reduction	free
λ_2 (extended multiplier)	fixed by the choice of section	$\nabla^{-2}\rho$
$A_0^{\text{eff}} = A_0 - \lambda_2$	**physical**; the Lagrangian A_0	Coulomb potential

Where that leaves us. The question is answered, but the route to the answer used a count that deserves scrutiny. Section 3.3.2 found one arbitrary function in H_T and two gauge conditions to impose with it, and concluded that only one condition may be chosen freely. Stated that way it sounds as though the Hamiltonian formalism is short of freedom by one function — which would be strange, since nothing was lost in passing from L to H. The next subsection shows that nothing is short, and that the correct statement is about the *kinds* of freedom available rather than their number.

3.5 Why one arbitrary function suffices

To write (one page, results only; proofs in Appendix A).

- Theorem A.3: complete gauge fixing needs N conditions, and their consistency equations have exactly N unknowns — the N_1 multipliers together with the $N - N_1$ coordinates further down the chains. Here $N = 2$, $N_1 = 1$, and the second unknown is the coordinate A_0 itself.
- Depth (Definition A.4): $\chi_1 = A_0$ has depth one, $\chi_2 = \nabla\cdot A$ depth two. This is why the Coulomb condition alone fixes the gauge completely with no extension of H_T (Proposition A.8), while the temporal condition alone leaves a time-independent $\alpha(\vec{x})$.
- So the count of §3.3.2 is not a shortage: the extended multiplier λ_2 of §3.3 is needed only to impose both conditions literally, and it is fixed by consistency, not a new degree of freedom.

3.6 The quantum side: one condition or two give the same theory

To write. Name the two procedures. (Q1) Fix χ_1 only, quantize (13) with the canonical commutator, impose Φ_2 on states as (14): quantize, then reduce. (Q2) Fix χ_1 and χ_2, quantize the Dirac bracket (26): reduce, then quantize. State the claim of the section: Q1 and Q2 are unitarily equivalent, so for this model quantization commutes with reduction, $[Q, R] = 0$, in the precise sense of Remark 3.26.

3.6.1 Gauss's law on states produces the Coulomb Hamiltonian

To write. Split $\hat{E} = \hat{E}^T + \hat{E}^L$. On states obeying (14), $\hat{E}^L = \nabla\nabla^{-2}\hat{\rho}$ (sign in the note's conventions to fix). Because $[\hat{\rho}(x), \hat{\rho}(y)] = 0$, the operator $\frac{1}{2}\int(\hat{E}^L)^2 = \frac{1}{2}\int \hat{\rho}\,(-\nabla^2)^{-1}\hat{\rho}$ needs no ordering choice. The dressing $U = \exp\bigl(ie\int \hat{\rho}\,\nabla^{-2}\nabla\cdot\hat{A}\bigr)$ (normalization to fix) maps the Gauss-law states of Q1 to wavefunctionals of (A^T, ϕ) and $\hat{H}_{\text{temporal}}$ to the Coulomb Hamiltonian (29), which is Q2. Zero mode: the total charge, §6.

Example 3.24 (two sites, one link)**.** Smallest lattice where both procedures can be written out: sites $1, 2$, link A, compact or noncompact. Write Q1 (Gauss's law at each site) and Q2 (eliminate the longitudinal part) and show the two Hamiltonians are unitarily equivalent. [Check to run.]

3.6.2 External charges

Static charges ρ_{ext}: the energy is the minimum of the kinetic energy subject to Gauss's law at fixed configuration (the amended potential of reduction at nonzero momentum). Without condensate this gives the Coulomb energy; with condensate $|\phi| = v$ it is screened.

Example 3.25 (Coulomb and Yukawa energy)**.** One longitudinal mode: minimize $\frac{1}{2}E_L^2 + \frac{1}{2}\pi^2$ subject to $kE_L - ev\,\pi = \rho$; the multiplier is the static A_0 and $K_{\min} = \rho^2/2(k^2 + M^2)$, $M = ev$; Fourier transform gives $q_1 q_2 e^{-Md}/4\pi d$. [Check C9.]

Remark 3.26 (what exactly commutes)**.** To write (one paragraph). Q2 agrees with Q1 because it is written in the linear variables (A^T, ϕ) and inherits the ordering of $\hat{\rho}(-\nabla^2)^{-1}\hat{\rho}$ from Q1. Quantizing the reduced space intrinsically, by the Laplace–Beltrami operator of its quotient metric, would differ from Q1 by the orbit-density term $\frac{\hbar^2}{2}\Delta\sqrt{v}/\sqrt{v}$, $v = \det{}^{1/2}(-\nabla^2 + e^2|\phi|^2)$. This is where the curvilinear unitary gauge of §5 needs care.

4 Why the lattice fixes only one

To write (half a page, bridge). Section 3 showed that fixing one condition or two gives the same quantum theory. Yet the lattice construction fixes only χ_1 and keeps Φ_2 as a condition on states. The question of this section: what property of the pair (Φ_1, χ_1), as opposed to (Φ_2, χ_2), makes it the one to eliminate? The answer (Interpretation 4.4) is neither primary versus secondary nor first versus second class, but whether the reduced algebra of observables is local. Revise the opening paragraph of §4.1, which still reads as a continuation of the chain analysis now in Appendix A.]

4.1 What survives, and what does not

Section A.1 answers the question it was set: nothing is short, and what distinguishes χ_1 from χ_2 is depth — 1 against 2 — hence the kind of resource each consumes. Behind that lies the double role of A_0, which by Lemma A.1 is at once the coefficient of Φ_2 in H_c and the coordinate conjugate to Φ_1.

But that answer carries an objection with it. Depth is measured against a chain anchored at the primary constraints, and it is not stable: under H_E every condition has depth one and the distinction evaporates. So depth is a property of the presentation of the theory, not of the theory. The question this subsection asks is whether anything about the difference between Φ_1 and Φ_2 survives.

4.1.1 The first candidate: primary versus secondary

Chain position is primary-versus-secondary in disguise, and that classification is not a property of the theory.

Example 4.1 (the same system with and without constraints)**.** $L = \frac{1}{2}\dot{q}^2$ has no constraints. The first-order presentation of the same system, $L' = p\dot{q} - \frac{1}{2}p^2$ with (q, p) both configuration variables, has $\pi_q = p$, $\pi_p = 0$, hence two *primary, second-class* constraints

$$\Phi_a = \pi_q - p \approx 0, \qquad \Phi_b = \pi_p \approx 0, \qquad \{\Phi_a, \Phi_b\} = -1, \tag{41}$$

with $H_c = \frac{1}{2}p^2$ and no secondary stage. The existence of constraints, their number, and their primary/secondary labels all depend on the Lagrangian representative; the reduced phase space (q, p) and its dynamics do not.

The split is a statement about the image of $\mathbb{F}L : TQ \to T^*Q$, hence about the vertical foliation of T^*Q, and a canonical transformation not preserving that foliation destroys the notion of "primary". First and second class refer instead to the degeneracy of $\omega|_C$ and carry no such dependence, which is why the usual advice is to remember the second classification and forget the first. The present model shows the point a second way: in the presentation of §2.3, with A_0 dropped at the outset, $\nabla\cdot E - \rho$ is not a constraint at all but a conserved quantity, imposed on states by (14) as a superselection choice. So depth, and with it everything in §A.1 that is phrased through the chain, belongs to the presentation.

4.1.2 The second candidate: first versus second class

The surviving classification cannot be the answer either, and for a blunt reason: in (1) both Φ_1 and Φ_2 are first class, so it does not distinguish them at all. It is worth seeing what happens when it is varied while everything else is held fixed.

Example 4.2 (Proca)**.** For $\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}m^2 A_\mu A^\mu$ the primary/secondary labels are word for word those of §2: $\Phi_1 = \pi^0 \approx 0$, and consistency yields $\Phi_2 = \nabla\cdot E + m^2 A_0 \approx 0$. But H_c now carries $-\frac{1}{2}m^2\int A_0^2$, so by (48) $\{\Phi_1, \Phi_2\} = -m^2 \neq 0$: the pair is *second* class, with no gauge freedom, no χ, no H_E, and three polarizations rather than two. Yet

$$\{A_i, E_j\}_D = \delta_{ij}, \qquad \{A_i, A_j\}_D = \{E_i, E_j\}_D = 0, \qquad \{A_0, \pi^0\}_D = 0 \tag{42}$$

hold exactly: the (A_0, π^0) sector is deleted with no correction to the physical brackets, exactly as in Proposition 2.5.

The labels agree between Maxwell and Proca while the physics does not, so the labels predict nothing. The class structure differs while the clean deletion of (A_0, π^0) happens in both, so that does not account for it either. Proposition A.2 says why the two can come apart: that $\dot{A}_0$ is absent from L, which makes A_0 auxiliary and its sector detachable, is a statement one derivative earlier than $\partial^2_{A_0} H_c$, which is what decides first versus second class. Proca holds the first fixed and flips the second.

So the feature to look for is whatever it is about A_0 that makes its sector detach — and the sharpest way to ask that is to ask when the Dirac bracket is needed at all.

4.1.3 What does survive

Proposition 4.3 (the Dirac bracket is a coordinate device)**.** *For second-class Ξ_a let*

$$\mathcal{A} := \big\{ f : \{f, \Xi_a\} = 0 \quad \forall a \big\} \tag{43}$$

be their Poisson commutant. Then $\{f, g\}_D = \{f, g\}$ for $f \in \mathcal{A}$ and every *g; $\mathcal{A}$ is a Poisson subalgebra; and $\iota^*|_{\mathcal{A}}$ is a Poisson homomorphism onto $C^\infty(\Sigma)$, surjectivity holding near Σ by Weinstein's normal form. The induced structure is therefore always computable with the ambient bracket alone, on variables drawn from $\mathcal{A}$.*

Proof. Invertibility of C makes the correction $\{f, \Xi_a\}(C^{-1})^{ab}\{\Xi_b, g\}$ vanish as soon as one argument lies in $\mathcal{A}$; the Jacobi identity gives closure. For surjectivity, a tubular neighbourhood of Σ is symplectically $\Sigma \times F$, and $\mathcal{A}$ contains the pullbacks along the projection. □

This disposes of a tempting way to state the difference. One might say that (Φ_1, χ_1) is special because it splits off a symplectic direct factor $\mathcal{P} = \mathcal{P}_{\text{phys}} \times T^*\mathbb{R}_{(A_0)}$ while (Φ_2, χ_2) does not; but by Weinstein every second-class pair splits off a direct factor locally, and by Proposition 4.3 the correction term is never needed in principle. It is the price of working outside $\mathcal{A}$. What differs between the two eliminations is which fields $\mathcal{A}$ contains.

For (Φ_1, χ_1) it contains the ones already in use, since $\{F, \pi^0\} = \{F, A_0\} = 0$ for every F built from $(A_i, E_i, \phi, \pi_\phi)$; the product is moreover global and exact rather than tubular. For (Φ_2, χ_2) the commutant is generated by the transverse fields instead, since $\mathsf{P}^T = \mathbf{1} - \nabla\nabla^{-2}\nabla\cdot$ annihilates gradients and divergences:

$$\{A_i^T, \Phi_2\} = \{A_i^T, \chi_2\} = \{E_i^T, \Phi_2\} = \{E_i^T, \chi_2\} = 0, \tag{44}$$

and there too the ambient bracket already returns the Dirac answer,

$$\{A_i^T, E_j^T\} = \mathsf{P}^T_{ik}\mathsf{P}^T_{jk} = \delta^T_{ij} = \{A_i, E_j\}_D, \tag{45}$$

by idempotence of the projector. No Dirac bracket is needed here either — but A^T and E^T are nonlocal functionals of A and E, so reaching $\mathcal{A}$ costs ∇^{-2}, and the correction $\delta_{ij} - \partial_i\nabla^{-2}\partial_j$ is exactly what one pays for keeping the local fields.

Interpretation 4.4 (what the asymmetry is)**.** Not primary versus secondary, which is not invariant (Example 4.1); not first versus second class, which does not distinguish the two here and does not track the phenomenon when varied (Example 4.2); but **whether the commutant** (43) **contains the local fields**. For (Φ_1, χ_1) it does, because $\dot{A}_0$ is absent from L and the constraint is algebraic in the fields; for (Φ_2, χ_2) it contains only their transverse projections, because Gauss's law is differential. The operative property of a constraint is its differential order, and what it controls is whether the reduced algebra of observables is local.

What is never true is $\{\ ,\ \}_D = \{\ ,\ \}$ on all of $C^\infty(\mathcal{P})$, which would force $C^{ab} = 0$. Indeed $\{A_0, \pi^0\}_D = 1 - 1 = 0 \neq \{A_0, \pi^0\}$ — the bracket doing its job, the eliminated pair becoming Casimir.

Remark 4.5 (consequence for the lattice)**.** This is why only Φ_2 appears on the lattice, and it is a fourth reason to add to the three given in the companion note *Toric Code Demystified*, §2.1. Proposition 2.5 lets (A_0, π^0) be discarded before discretizing with the canonical bracket carried across untouched, so $[\hat{A}_i, \hat{E}_j] = i\delta_{ij}$ survives verbatim and $\hat{E}_j = -i\,\delta/\delta A_j$ remains available — which the smearing onto links in the companion note relies on. Eliminating (Φ_2, χ_2) instead would require representing $[\hat{A}_i, \hat{E}_j] = i\delta^T_{ij}$, for which the naive functional derivative is unavailable. Locality of the reduced bracket, not the classification of the constraints, is what selects the temporal presentation of §2.3 as the one to discretize.

4.2 From the continuum to the lattice

To write (one page). Collect what the companion note uses: (13) with (14); the canonical commutator surviving the elimination of (π^0, A_0) (Proposition 2.5); smearing onto links, A_ℓ with conjugate E_ℓ; Gauss's law as an oriented lattice divergence at each site [KS75]. Point forward to §6 for the step from $\mathbb{R}$ to $U(1)$ link variables, which is a choice, not a consequence.

5 Unitary gauge and the Higgs phase

Role: a third choice of the second gauge condition, local like temporal gauge and complete like Coulomb gauge, and the gauge in which the Higgs mechanism is usually described. The section asks what that description does and does not mean.

5.1 Unitary gauge as a third choice of χ_2

To write. $\chi_2 = \theta$, the phase of $\phi = re^{i\theta}$; $\{\chi_2, \Phi_2\}$ is invertible wherever $r \neq 0$. Reduced variables $A' = A - \nabla\theta/e$ and r; the reduced Hamiltonian is a massive vector (Proca-like, mass er) coupled to the radial mode, and its brackets are local — contrast with δ^T_{ij} of Coulomb gauge. Price: the condition is singular at zeros of ϕ. Relate to the Proca example of §4.1.

5.2 What is gauge invariant

To write. Elitzur's theorem [Eli75]: without gauge fixing, $\langle\phi\rangle = 0$. In Coulomb gauge $\langle\phi\rangle$ can be nonzero, because there ϕ is the Dirac-dressed field $e^{ie\nabla^{-2}\nabla\cdot A}\phi$ [Dir55], gauge invariant but nonlocal (for the noncompact lattice model see [KK86], to check). So the Higgs phase is not a broken gauge symmetry; what Q1 and Q2 share is the spectrum, not the locality of their variables. Pointer to the companion note's discussion of order parameters and [FS79].

Example 5.1 (a dressed operator on two sites)**.** On the two-site lattice of Example 3.24: write the dressed matter operator, check that it commutes with Gauss's law at both sites, and show its weight on the link. [Check to run.]

Remark 5.2 (measure and the singular slice)**.** To write (one paragraph). Unitary-gauge variables are curvilinear, so quantizing the reduced Hamiltonian requires a choice of measure; the one that agrees with Q1 includes the orbit-density term of Remark 3.26. In the zero-dimensional model it is exactly $V_\rho = -\frac{e^2(e^2r^2-4)}{8(1+e^2r^2)^2}$, which tends to $-1/8r^2$ for $er \gg 1$ (the unitary-gauge Jacobian $\prod_x |\phi_x|$) and is finite at $r = 0$: the singularity at zeros of ϕ belongs to the slice, not to the reduced space. [Check C10.]

6 Compact gauge group and holonomies

Role: the third assumption of the companion note, and the bridge to it.

6.1 Charge quantization

To write. With $\alpha(\vec{x})$ periodic (compact $U(1)$), Gauss's law has integer eigenvalues and charge is quantized; on links e^{iA_ℓ} and integer E_ℓ. In the noncompact theory nothing forces this. One remark on the static potential: continuum Coulomb (or Yukawa, Example 3.25) versus the linear potential of electric flux strings at strong coupling on the compact lattice.

6.2 Large gauge transformations on a circle

To write. On a closed space Gauss's law constrains only small gauge transformations; the constant ones force total charge zero (the zero mode of ∇^2 in §3), and the winding ones act on the holonomy, which Coulomb gauge leaves unfixed.

Example 6.1 (an N-site ring)**.** Compact $U(1)$ on a ring of N sites: reduce by Gauss's law, isolate the holonomy mode, and compute its spectrum (including a possible θ-like shift) and the total-charge constraint. [Check to run.]

6.3 Towards the toric code

To write (half a page). Reduce $U(1)$ to $\mathbb{Z}_2$; holonomies on a torus give the four-fold degeneracy; pointer to *Toric Code Demystified* §2–§3 [Kit03].

A Constraint chains and depth

[One-paragraph introduction to write: this appendix proves the statements summarized in §3.5. Its first subsection was written as a continuation of §3; adjust its opening sentence.]

A.1 Chains, and how gauge conditions are absorbed

Write $\Phi_1, \dots, \Phi_N$ for the first-class constraints, N_1 of them primary, and fix signs by

$$\Phi_{k+1} := \{\Phi_k, H_c\}, \tag{46}$$

which for (7) gives $\Phi_2 = \{\pi^0, H_c\} = \nabla{\cdot}E - \rho$, the constraint of Proposition 2.2. Since ad_{H_c} is nilpotent on the first-class set, the constraints organise into chains $\Phi_1 \to \cdots \to \Phi_L \to 0$, one per primary.

A.1.1 What the algorithm does to H_c

Lemma A.1 (chain lemma)**.** *Let $\Phi_1 = p_{y_1} \approx 0$ be primary because $\dot{y}_1$ is absent from L. Then*

$$H_c = H_0 - \sum_{k=1}^{L-1} y_k\, \Phi_{k+1}, \qquad \{y_k, \Phi_j\} = \delta_{kj}, \tag{47}$$

with H_0 independent of every y_k: the coefficient of Φ_{k+1} in H_c is the coordinate conjugate to Φ_k, one step back along the chain.

Proof. $\Phi_2 = -\partial H_c/\partial y_1$, and for Φ_2 to be a constraint independent of y_1 the Hamiltonian must be affine in y_1, so $H_c = H_0^{(1)} - y_1\Phi_2$. Repeat on the coordinate conjugate to Φ_2, and so on until $\Phi_{L+1} = 0$. □

For $L = 2$ this is what §3.3.2 observed: in (7) the coefficient of Φ_2 is $y_1 = A_0$, conjugate to $\Phi_1 = \pi^0$. It is the reason A_0 was pulled two ways in §3.3, and (47) says the coincidence is structural rather than a feature of Maxwell.

The affineness used in the proof is not an extra hypothesis. It is what first class means:

Proposition A.2 (why a chain continues, or stops)**.** *With $\Phi_1 = p_{y_1}$ and $\Phi_2 = -\partial H_c/\partial y_1$,*

$$\{\Phi_1, \Phi_2\} = \frac{\partial^2 H_c}{\partial y_1^2}. \tag{48}$$

So H_c affine in y_1 $\iff$ first class, the chain continuing with y_1 free; H_c nondegenerate quadratic in y_1 $\iff$ second class, the chain stopping and $\Phi_2 = 0$ determining y_1.

For (1), A_0 enters H_c linearly, $\partial^2_{A_0} H_c = 0$, and the chain continues — this is gauge invariance. Adding a mass term, as in Example 4.2 below, gives $\partial^2_{A_0} H_c = -m^2$, and $\Phi_2 = 0$ reads $A_0 = -m^{-2}\nabla{\cdot}E$ instead. Gauge invariance and its absence are the two branches of one derivative, a fact §4.1 will need.

A.1.2 There is no shortage

Theorem A.3. *Let the first-class constraints form N_1 chains of lengths L_a, so $N = \sum_a L_a$. Complete gauge fixing needs N conditions χ_A, and their consistency*

$$\dot{\chi}_A = \{\chi_A, H_c\} + \sum_{a=1}^{N_1} \lambda_a \{\chi_A, \Phi_1^{(a)}\} = 0 \tag{49}$$

is N equations in exactly N unknowns: the multipliers $\lambda_1, \dots, \lambda_{N_1}$, which are arbitrary functions of time, together with the $N - N_1$ phase-space coordinates $y_k^{(a)}$, $k = 1, \dots, L_a - 1$.

Proof. The non-multiplier unknowns enter through $\{\chi_A, H_c\}$, whose only y-dependence is the $\sum_k y_k \Phi_{k+1}$ term of (47), giving $L_a - 1$ of them per chain; and $N_1 + \sum_a (L_a - 1) = N$. □

So the count is never short, and §3.3.2 was reporting something else: the two kinds of unknown are not interchangeable. A multiplier may be assigned at will; determining a y_k restricts the state to a surface — legitimate, since y_k is conjugate to a first-class constraint and hence pure gauge, but a different act. Which condition draws on which kind is therefore the thing to understand.

A.1.3 Depth

Definition A.4. The *chain depth* of a gauge condition χ is the least m such that $\partial_t^m \chi$, computed with H_T, contains an undetermined multiplier.

The definition mentions only χ and H_T. Evaluating it produces the chain of its own accord:

Proposition A.5. *The coefficient of λ_a in $\partial_t^m \chi$ is $(-1)^{m-1}\{\chi, \Phi_m^{(a)}\}$ whenever the lower ones vanish. Hence*

$$\operatorname{depth}(\chi) = \min\big\{k : \{\chi, \Phi_k\} \neq 0\big\}. \tag{50}$$

Proof. $\dot\chi = \{\chi, H_c\} + \lambda\{\chi, \Phi_1\}$ gives $m = 1$. If $\{\chi, \Phi_1\} = 0$ then by Jacobi

$$\{\{\chi, H_c\}, \Phi_1\} = \underbrace{\{\{\chi, \Phi_1\}, H_c\}}_{0} - \{\chi, \underbrace{\{\Phi_1, H_c\}}_{\Phi_2}\} = -\{\chi, \Phi_2\}, \tag{51}$$

which is the induction step; iterate. □

Each differentiation moves ad_{H_c} off χ and onto Φ_1, which is why the chain appears at all: it is generated by the induction rather than assumed. That also fixes what (50) may be applied to. The Φ_k there are not an arbitrary basis of the first-class set but the ladder grown from a *primary* constraint, and the primary set is fixed by the image of $\mathbb{F}L$, not chosen. Anchoring elsewhere breaks the formula: in the three-link model below, $p_3 + p_1$ is first class but not primary, and reading (50) against the ladder it generates assigns depth 1 to a condition whose true depth is 3.

Remark A.6 (depth does not depend on the other conditions)**.** Definition A.4 has two arguments, χ and H_T, and $H_T = H_c + \lambda_a \Phi_1^{(a)}$ contains no gauge condition. So the depth of χ_A is unchanged by which other conditions are imposed, and in particular by any order among them; there is no order in (49) to depend on, since it is one simultaneous system. This must not be confused with a different procedure: if one *reduces* after each condition — passing to the Dirac bracket and continuing on the smaller phase space — then H_T is replaced and the chain does change. Fixing $\chi_1 = A_0 - v$ and reducing leaves $H_{\mathrm{red}} = H_0 - \langle v, \Phi_2\rangle$ with $\{\Phi_2, H_{\mathrm{red}}\} = 0$, a chain of length one and no multiplier. That is depth measured in a different presentation, not order dependence, and the two routes agree on every physical conclusion, as Remark 3.17 requires: both give $\{\chi_2, H\} = \nabla\cdot E + \nabla^2 v$, which vanishes for $v = -\nabla^{-2}\rho$ and not for $v = 0$.

Proposition A.7 (triangularity)**.** *Write w_k for the coordinate conjugate to Φ_k, so $w_k = y_k$ for $k < L$ while w_L is not a resource. A depth-k condition is a condition on w_k, and $\dot\chi = 0$ determines w_{k-1}, with $w_0 := \lambda$.*

Each condition is absorbed by the resource one step *back* along its chain: the one at the top consumes the multiplier, every one below consumes a coordinate. The depths within a chain are $1, \ldots, L_a$, all distinct, so no two conditions compete and (49) closes triangularly.

The chain does have a direction, though not one of ordering. Imposed *alone*, a depth-k condition propagates upward: $\dot{\chi} = 0$ gives w_{k-1}, $\ddot{\chi} = 0$ gives w_{k-2}, and so on to λ, consuming k resources. A condition at the bottom of a chain therefore fixes that chain completely by itself, while one at the top fixes only λ and leaves a residual. For (1) this is the asymmetry §3.3 ran into, stated without reference to H_E:

Proposition A.8. *The Coulomb condition $\chi_2 = \nabla \cdot A \approx 0$, imposed alone, is preserved by the total Hamiltonian (8) and fixes the gauge completely, with no multiplier for Φ_2 and no modification of H_T. The temporal condition $\chi_1 = A_0 \approx 0$, imposed alone, does not: it leaves a time-independent $\alpha(\vec{x})$.*

Proof. χ_2 has depth two, so $\dot{\chi}_2 = \nabla \cdot E + \nabla^2 A_0 = 0$ determines $w_1 = A_0$ at $-\nabla^{-2}\rho$ and $\ddot{\chi}_2 = 0$ then gives $\lambda_1 = -\nabla^{-2}\dot{\rho}$; both resources are consumed. χ_1 has depth one and consumes only λ_1. □

A.1.4 Where a set of conditions can fail

A depth-k condition speaks about w_k twice over: it prescribes a value outright, and the consistency of the depth-$(k{+}1)$ condition derives one. The two must agree.

Proposition A.9 (compatibility)**.** *Impose one condition of each depth $k = 1, \ldots, L$ on a chain, the depth-k one prescribing $w_k = v_k$, and let $\hat{w}_k$ denote the value that consistency of the depth-$(k{+}1)$ condition derives. The system is solvable iff*

$$v_k = \hat{w}_k, \qquad k = 1, \ldots, L-1. \tag{52}$$

Only the bottom prescription v_L and the top multiplier λ are unconstrained.

For $L = 2$ there is exactly one such condition, and it is Theorem 3.22. That theorem is therefore not a fact about Maxwell but the $L = 2$ instance of (52): the A_0 prescribed by χ_1 must equal the $-\nabla^{-2}\rho$ that $\dot{\chi}_2 = 0$ derives. The choice $\chi_1 = A_0$ prescribes 0 and fails once $\rho \neq 0$; the choice $\chi_1 = A_0 + \nabla^{-2}\rho$ prescribes exactly $\hat{w}_1$ and passes. Both readings of the failure — that χ_2 has already claimed A_0, or that $\dot{\chi}_2$ has no unknown left — are the same equation, which is symmetric in the two conditions.

A.1.5 What is actually being chosen

Equation (52) is easier to use if the conditions are written with their prescribed values displayed. At $L = 2$, put $\chi_1 = A_0 - v$ and $\chi_2 = \nabla \cdot A - u$.

Corollary A.10. *The pair is admissible under H_T iff*

$$\dot{u} - \nabla^2 v = \rho, \tag{53}$$

which is (40) with $v = A_0^{\text{eff}}$ and $u = \nabla \cdot A$. Hence a complete gauge fixing is a choice of one *of the following, the other being determined:*

prescribe	*determined*	*example*
$u(\vec{x}, t)$	v, *pointwise in* t	*Coulomb:* $u = 0 \Rightarrow \nabla^2 v = -\rho$
$v(\vec{x}, t)$	$\dot{u}$, *leaving* $u_0 = u(\vec{x}, 0)$ *free*	*temporal:* $v = 0 \Rightarrow \dot{u} = \rho$

The two rows carry the same information: u decomposes into u_0 and $\dot{u}$, so prescribing u and prescribing (v, u_0) are choices of the same size. What one trades is which function is determined, not how many.

This settles a question the temporal gauge leaves open. After $\chi_1 = A_0 - v$ the residual gauge freedom is a time-independent $\alpha(\vec{x})$, one function of *space*. By Corollary A.10 the freedom still available in χ_2 is $u_0(\vec{x})$, also one function of space — and it is the same one, since a residual transformation shifts $\nabla\cdot A$ by $-\nabla^2\alpha$ at $t = 0$. So the residual can be removed exactly, with no extension of H_T, by prescribing u_0 and letting $\dot{u}$ be what (53) says. On the lattice, with v chosen so that $\rho + \nabla^2 v$ has magnitude 9.89, evolution under H_T keeps $\nabla \cdot A(t) = u_0 + t\,(\rho + \nabla^2 v)$ to 5×10^{-13} out to $t = 2$; changing u_0 alone leaves $\vec{E}$ and $\vec{B}$ unchanged to 2×10^{-13} and shifts $\vec{A}$ by a pure gradient.

What one cannot do is prescribe v and u independently, which is one function of spacetime too many; radiation gauge, $u \equiv 0$ with $v \equiv 0$, is the case in point. This is the general role of the extended Hamiltonian. Attaching a multiplier to every Φ_k makes $\dot{\chi} = 0$ determine λ_k rather than w_{k-1}, so every condition has depth one, no condition's consistency reaches a coordinate another has prescribed, and all $L - 1$ instances of (52) evaporate. H_E buys no freedom — Theorem A.3 keeps the total at N — it buys the deletion of (52), at the price recorded in Proposition 3.12.

A.1.6 Why $N_1 < N$

Proposition A.11. *$N_1 \le N$, with equality iff every chain has length one. In general $L_a = 1 + \max\{k : \epsilon_a^{(k)}$ appears in $\delta q\}$, and*

$$G[\epsilon] = \sum_a \sum_{k=0}^{L_a - 1} (-1)^k\, \epsilon_a^{(k)}\, \Phi^{(a)}_{L_a - k}. \tag{54}$$

The alternating sign is forced by (46): for (1), $G[\epsilon] = \int(\epsilon\Phi_2 - \dot{\epsilon}\Phi_1)$ generates $\delta A_\mu = -\partial_\mu\epsilon$ and $\delta\phi = -ie\epsilon\phi$ uniformly, whereas the opposite relative sign gives $\delta A_0 = +\dot{\epsilon}$ against $\delta A_i = -\partial_i\epsilon$, which is not a gauge transformation of anything.

Here is why the two counts differ. A gauge transformation is N_1 arbitrary functions *of time*, one per chain; but phase space at a fixed instant sees not ϵ_a alone, rather its $(L_a - 1)$-jet, since $\epsilon_a, \dot{\epsilon}_a, \ldots$ may be prescribed independently at t_0. The orbit through a point therefore has dimension $\sum_a L_a = N$. The gap $N - N_1$ is not a deficit of freedom but the difference between counting a symmetry as functions of time and as directions on initial data, each lengthening of a chain being one differentiation of ϵ. It is the same distinction that appeared in Corollary A.10, where the leftover freedom was a function of space rather than of spacetime.

Remark A.12 (a chain of length three)**.** $L = \frac{1}{2}(\dot{q}_1 - q_2)^2 + \frac{1}{2}(\dot{q}_2 - q_3)^2$ is invariant under $\delta q_1 = \epsilon$, $\delta q_2 = \dot{\epsilon}$, $\delta q_3 = \ddot{\epsilon}$. Only $\dot{q}_3$ is absent, so

$$H_c = \tfrac{1}{2}p_1^2 + \tfrac{1}{2}p_2^2 + q_3 p_2 + q_2 p_1, \qquad \Phi_1 = p_3 \to \Phi_2 = -p_2 \to \Phi_3 = p_1 \to 0, \tag{55}$$

all first class: $N = 3$, $N_1 = 1$, and $y_1 = q_3$, $y_2 = -q_2$ by (47). Conditions on q_3, q_2, q_1 have depths $1, 2, 3$, their consistencies being λ, $p_2 + q_3$ and $p_1 + q_2$, so they determine λ, w_1 and w_2 and leave the two instances of (52) that q_3 and q_2 be prescribed at $-p_2$ and $-p_1$. Formula (54) reads $G = \epsilon p_1 + \dot{\epsilon} p_2 + \ddot{\epsilon} p_3$, reproducing $\delta q_k = \epsilon^{(k-1)}$.

The numbers above are model-specific; the mechanism is not. $N_1 = N$ does occur — if L is independent of a coordinate y altogether then $\{p_y, H_c\} = 0$ terminates that chain at length one and $H_T = H_E$ — so $N_1 < N$ is no theorem. What is general is (47), (48), Theorem A.3 and (52). For Maxwell, Yang–Mills and general relativity the gauge parameter enters δq with at most one time derivative, so every chain has length two and $N_1 = N/2$.

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